Platform Modules Architecture & Operational Blueprint

Complete Technical Specification of Platform Micro-Modules, Data Pipelines, and Security Boundaries

TOTAL BACKEND MODULES	FRONTEND ENGINES	DATABASE LAYER	RUNTIME CLUSTER
55+ Integrated Modules	React Native Web (Expo)	Neon PostgreSQL + Prisma	PM2 + Linux Cloud VPS

1. HIGH-LEVEL ARCHITECTURAL TOPOLOGY & COMPONENT INTERACTION

SYSTEM TOPOLOGY

The platform operates on a modular, decoupled tier topology. Client and Administrative frontends interact with the NestJS gateway over SSL/TLS encrypted REST and WebSockets. Incoming requests pass through automated rate limiters, JWT authorization guards, and Prisma database interceptors before reaching micro-service modules.

1. Client Web View
React Native Web (Port 8081). Farmer, Gardener & Customer workflows, responsive 60 FPS transitions.

2. Admin Console
React Native Web (Port 8082). Executive overview, staff RBAC desks, and financial clearing consoles.

3. Core API Gateway
NestJS Modular Server (Port 3000). Throttler guards, helmet security, and business micro-services.

4. Database & Cache
Neon PostgreSQL Serverless pool with Redis fast session cache and immutable audit tables.

2. MODULE CLUSTER 1: CORE GOVERNANCE, SECURITY & IDENTITY

GOVERNANCE & AUTH

This foundational cluster governs authentication, user identity lifecycles, cryptographic token verification, and statutory ISO compliance across the platform.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS
AuthModule SECURITY CORE	Issues cryptographic JWT tokens, handles mobile OTP carrier dispatch, password bcrypt hashing, and token refreshes.	Receives credentials from Client/Admin, validates against `UsersModule`, injects user context into JWT claims.	`POST /auth/login` `POST /auth/verify-otp` `POST /auth/refresh-token`
UsersModule PROFILE MANAGEMENT	Manages multi-tier user entity records, addresses, KYC verification flags, landholding records, and role states.	Provides user hydration to `FarmsModule`, `WalletModule`, and partner relationship managers.	`GET /users/me` `PATCH /users/profile` `GET /users/kyc-status`
AuditModule ISO 27001 AUDIT	Interceptors automatically record operator ID, IP address, timestamp, and before/after payloads for every write mutation.	Hooks into `AuditLogInterceptor`, writing immutable records to PostgreSQL without blocking client responses.	`GET /audit/logs` `GET /audit/trail/:userId` `POST /audit/export`
ComplianceModule STATUTORY GUARD	Enforces statutory e-MoA objects, terms of service synchronization, corporate privacy policies, and refund rules.	Supplies legally binding corporate disclosures to frontend compliance views and public legal footers.	`GET /compliance/policies` `GET /compliance/statutory` `POST /compliance/consent`
AppSettingsModule DYNAMIC CONTROLS	Provides real-time feature flag toggles, maintenance mode triggers, advisory rate cards, and global system variables.	Queried at frontend app bootstrap to dynamically show/hide diagnostic and transactional capabilities.	`GET /app-settings` `PATCH /app-settings/toggle` `GET /app-settings/public`

3. MODULE CLUSTER 2: AGRONOMY INTELLIGENCE & AI CROP DOCTOR

AGRONOMY AI

This subsystem powers the AI Leaf Scanner, pest classification models, certified dosage calculations, and automated crop activity scheduling aligned with Indian Council of Agricultural Research (ICAR) agronomy protocols.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS
CropProblemsModule AI DIAGNOSTIC CORE	Computer vision classification of leaf diseases, fungi, pests, and nutrient deficiencies with confidence scoring.	Consumes image uploads from `UploadsModule`, matches pathology with `SprayScheduleModule` templates.	`POST /crop-problems/diagnose` `GET /crop-problems/history` `GET /crop-problems/:id`
SprayScheduleModule DOSAGE CALCULATOR	Calculates exact chemical formulations, water mixing ratios, and application frequencies based on acreage.	Queries `SprayItemTemplatesModule` for approved active ingredients; sends reorder links to `ProductsModule`.	`POST /spray-schedules/calculate` `GET /spray-schedules/crop/:id` `PATCH /spray-schedules/complete`
CropActivitySchedules CROP LIFECYCLE	Automates multi-week agronomic timelines (sowing, basal fertilization, irrigation, weeding, and harvest windows).	Binds to plots in `PlotsModule`; triggers milestone push alerts via `NotificationsModule`.	`GET /crop-activity/timeline` `POST /crop-activity/custom-task` `PATCH /crop-activity/mark-done`
Farms & Plots Modules LAND PARCEL GRID	Tracks farm physical boundaries, geo-coordinates, soil classifications, plot dimensions, and primary cultivars.	Feeds geo-coordinates to `SatelliteModule` for NDVI bounding boxes and `WeatherModule` for station locks.	`GET /farms` `POST /farms/:id/plots` `PATCH /plots/:id/geo-lock`

4. MODULE CLUSTER 3: SPACE & ATMOSPHERIC TELEMETRY

SPACE & WEATHER

Combines orbital satellite remote sensing with hyper-local weather forecasting to advise farmers on optimal spray windows and detect vegetative moisture stress before visual symptoms appear.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS
SatelliteModule ORBITAL REMOTE SENSING	Processes Sentinel-2 and Landsat multispectral imagery to calculate Normalized Difference Vegetation Index (NDVI).	Reads polygon coordinates from `PlotsModule`; maps crop chlorophyll density and moisture gradients.	`GET /satellite/ndvi/:plotId` `GET /satellite/moisture-index` `POST /satellite/sync-tile`
WeatherModule ATMOSPHERIC TELEMETRY	Integrates Open-Meteo API for real-time temperature, relative humidity, wind speed, precipitation probability, and frost alerts.	Evaluates environmental safety before `SprayScheduleModule` recommends spraying (e.g. aborts if rain >60% or wind >15 km/h).	`GET /weather/forecast` `GET /weather/alerts` `GET /weather/spray-suitability`

5. DIAGNOSTIC PIPELINE: END-TO-END EXECUTION FLOW

WORKFLOW SEQUENCE

1. Leaf Scan Capture
Farmer captures affected crop leaf photo; `UploadsModule` stores encrypted image.

2. AI Disease Match
`CropProblemsModule` analyzes image, identifies pathogen (e.g. Yellow Rust), scores 98.4%.

3. Dosage Synthesis
`SprayScheduleModule` generates tank mix: Propiconazole 25% EC @ 200ml per 200L water/acre.

4. Weather Safety Lock
`WeatherModule` verifies 0% rain probability and optimal wind speed before confirming spray window.

6. MODULE CLUSTER 4: REAL-TIME MANDI PRICE DISCOVERY & FORECASTING

MARKET INTELLIGENCE

This cluster aggregates real-time agricultural commodity prices across hundreds of APMC mandis, identifies abnormal market fluctuations, and projects 7-day predictive price trends using machine learning regression.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS

MarketRatesModule
APMC RATE ENGINE
	Ingests, normalizes, and publishes daily mandi rates (Min, Modal, Max prices) and commodity arrival volumes.	Indexed by district coordinates; automatically delivers hyper-local commodity rates to Client and Partner home screens.	`GET /market-rates/live`
			`GET /market-rates/commodity/:name`
			`POST /market-rates/publish`

MandiAIModule
PRICE TREND PREDICTION
	Analyzes multi-year historical arrival trends, weather disruptions, and demand shifts to project 7-day price trajectories.	Consumes historical rate streams from `MarketRatesModule`; highlights profitable selling windows for farmers.	`GET /mandi-ai/predictions`
			`GET /mandi-ai/volatility-index`
			`GET /mandi-ai/recommendations`

7. MODULE CLUSTER 5: AGRI-COMMERCE, RETAIL INVENTORY & LOGISTICS

AGRI-COMMERCE & 3PL

Governs the end-to-end e-commerce supply chain for certified seeds, bio-fertilizers, and farm equipment, syncing inventory across local franchise stores and managing third-party logistics.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS

ProductsModule
CATALOG ENGINE
	Manages certified product SKUs, chemical compositions, pricing, technical specifications, and category taxonomies.	Supplies product listings to Client E-Commerce shop; linked to `SprayScheduleModule` for prescription fulfillment.	`GET /products`
			`GET /products/:id`
			`POST /products/create`

SellerModule
FRANCHISE INVENTORY
	Enables local Agri-Stores to sync physical warehouse stock, configure localized pricing, and manage retail store POS slips.	Allocates customer orders to nearest geo-located franchise store; credits trade commissions to `WalletModule`.	`GET /seller/inventory`
			`PATCH /seller/stock-update`
			`GET /seller/payout-summary`

OrdersModule
ORDER STATE MACHINE
	Orchestrates order state lifecycle (Created → Confirmed → Dispatched → In-Transit → Delivered → Settled).	Initiates payment intents via `RazorpayModule`; on payment success, dispatches courier manifest to `ShiprocketModule`.	`POST /orders/checkout`
			`GET /orders/my-orders`
			`PATCH /orders/:id/status`

ShiprocketModule
3PL COURIER INTEGRATION
	Automates shipment booking, generates printable Air Waybills (AWBs), and ingests live courier tracking webhooks.	Receives confirmed shipments from `OrdersModule`; updates customer real-time tracking screens automatically.	`POST /shiprocket/create-shipment`
			`GET /shiprocket/track/:awb`
			`POST /shiprocket/webhook`

8. COMMERCE & LOGISTICS INTEGRATION PIPELINE

COMMERCE FLOW

1. Cart & Checkout
Farmer adds prescribed fungicide; `OrdersModule` locks stock and calculates delivery taxes.

2. Payment Gateway
Instant UPI/Card payment verified via `RazorpayModule` webhook; order confirmed.

3. Franchise Dispatch
Local store alerted; `ShiprocketModule` generates AWB shipping label and schedules courier pickup.

4. Doorstep Delivery
e-POD verified upon handover; store retail margin credited immediately to `WalletModule`.

9. MODULE CLUSTER 6: MULTI-RAIL PAYMENT GATEWAYS

PAYMENT RAILS

To ensure high transaction success rates and zero downtime across rural payment networks, FarmsKing implements a multi-rail payment fallback architecture supporting Razorpay, Cashfree, and PhonePe.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS

RazorpayModule PRIMARY PAYMENT RAIL	Processes dynamic UPI QR codes, debit/credit cards, net banking, and automatic recurring plan subscriptions.	Cryptographically validates webhook signatures (SHA-256); triggers state transitions in `OrdersModule` and `PlanPaymentsModule`.	`POST /razorpay/create-order`
			`POST /razorpay/webhook`
			`POST /razorpay/verify-payment`

CashfreeModule SECONDARY FALLBACK RAIL	Instant payment fallback rail and high-speed auto-split payment disbursal for franchise commissions.	Automatically engages if primary gateway latency exceeds 2500ms; executes batch payouts directly to vendor accounts.	`POST /cashfree/order`
			`POST /cashfree/webhook`
			`POST /cashfree/payout`

PhonePeModule UPI DIRECT GATEWAY	Direct UPI Intent flow for native mobile transactions, minimizing checkout drop-offs in rural connectivity zones.	Transmits verified server-to-server transaction status callbacks to `FarmerPlanPaymentsModule`.	`POST /phonepe/initiate`
			`POST /phonepe/callback`
			`GET /phonepe/status/:id`

10. MODULE CLUSTER 7: WALLETS, LEDGER KHATA & DISBURSALS

WALLETS & INVOICING

Governs internal double-entry ledger balances, partner commission wallets, withdrawal approvals, digital membership vouchers, and statutory GST-compliant printable bills.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS

WalletModule DIGITAL PASSBOOK	Maintains real-time user credit/debit balances, advisory earnings, referral bonuses, and digital cash deposits.	Accumulates earnings from completed consultations and orders; supplies funds for instant withdrawals.	`GET /wallet/balance`
			`GET /wallet/transactions`
			`POST /wallet/deposit`

WithdrawalsModule BANK SETTLEMENT	Processes partner payout requests, verifies bank account details, and executes automated or manual bank transfers.	Debits `WalletModule` balance on request submission; creates tamper-proof audit records in `AuditModule`.	`POST /withdrawals/request`
			`GET /withdrawals/history`
			`PATCH /withdrawals/:id/approve`

Parties & SaleBills APMC & GST BILLING	Generates legal GST sales bills, mandi cess deductions, farmer khata ledger statements, and A4 printable invoice slips.	Calculates CGST/SGST breakdowns; attaches printable slips to completed e-commerce orders and APMC mandi trades.	`POST /sale-bills/generate`
			`GET /sale-bills/:id/pdf`
			`GET /parties/ledger/:id`

Subscriptions & Plans FARMER MEMBERSHIPS	Administers tiered farmer and gardener plans (Basic, Standard, King Premium), renewal reminders, and discount coupons.	Activates priority tele-consultations and free soil lab diagnostics upon payment verification.	`GET /farmer-plans`
			`POST /farmer-plans/subscribe`
			`POST /coupons/apply`

11. FINANCIAL SECURITY & RECONCILIATION STANDARD

SECURITY & ESCROW

Double-Entry Ledger Integrity

Every transaction creates reciprocal debit and credit entries. Wallet balances cannot be modified without matching immutable payment receipts or bank transaction references.

Automated Webhook Anti-Spoofing

Payment callbacks require cryptographic signature validation using secret HMAC keys. Unsigned or delayed callbacks are discarded to prevent replay attacks.

12. MODULE CLUSTER 8: OMNICHANNEL AI VOICE & COMMUNICATION

OMNICHANNEL AI

Enables seamless farmer engagement across multiple communication channels, including dial-in voice assistance, real-time WebSocket advisory chats, and automated WhatsApp alert broadcasts.

MODULE NAME	TECHNICAL FUNCTION & PURPOSE	INTER-MODULE DATA FLOW	PRIMARY ENDPOINTS / HANDLERS
VoiceAIModule SPEECH DIAGNOSTIC	Processes spoken farmer inquiries via speech-to-text, queries agronomic LLM models, and returns synthesized audio guidance.	Integrates with `CropProblemsModule` to provide verbal crop advice in clear English and simple terminology.	`POST /voice-ai/stream` `POST /voice-ai/synthesize` `GET /voice-ai/telemetry`
Chat & AdminChat REAL-TIME MESSAGING	Real-time WebSockets / SSE consultation desk linking farmers with verified Agronomy Advisors and customer support staff.	Stores permanent audit logs of all recommendations; triggers push notifications for unread messages.	`WS /chat/socket` `GET /chat/threads` `POST /chat/message`
WhatsappBotModule META CLOUD API	Dispatches automated daily mandi rate digests, leaf scan results, weather warnings, and order tracking links via WhatsApp.	Subscribes to events from `OrdersModule` and `MarketRatesModule` for instant carrier notification.	`POST /whatsapp/send-template` `POST /whatsapp/webhook` `POST /whatsapp/broadcast`
Notifications & SMS PUSH & CARRIER SMS	Multi-channel push notification engine (Firebase Cloud Messaging) and DLT-registered carrier transactional SMS gateway.	Sends OTPs for `AuthModule`; dispatches status updates for `WithdrawalsModule` and `OrdersModule`.	`POST /notifications/push` `POST /sms/send-otp` `GET /notifications/unread`

13. MODULE CLUSTER 9: AUTONOMOUS REAL-TIME SEO & GEO ENGINE

AUTONOMOUS SEO

The dedicated `SeoModule` operates autonomously in the background to ensure every newly added feature, mandi rate, and diagnostic capability is indexed across search engines and AI knowledge bots.

IndexNow Real-Time Pinger — ACTIVE

Submits 25 canonical platform URLs to IndexNow (Bing, Yandex) every 12 hours and 15 seconds after server bootstrap using the verified 32-char hexadecimal key (`7c23f4c12cd8c5dcddf6d149b8339b9f`).

Generative Engine Profiles (GEO) — ACTIVE

Generates and hosts machine-readable AI knowledge profiles (`llms.txt`, `llms-full.txt`, `ai-knowledge.json`) detailing all 17 capabilities for LLM crawlers (ChatGPT, Claude, Perplexity, Gemini).

14. MODULE HEALTH TELEMETRY & STATUTORY AUDIT CERTIFICATION

SYSTEM HEALTH

Backend Micro-Services — HEALTHY

55+ NestJS modules online; zero unhandled promise rejections; sub-45ms latency.

Database Pool — CONNECTED

Neon PostgreSQL SSL pooled connections; 24 verified accounts; clean schema.

SEO Background Runner — SCHEDULED

Automatic recurring 12-hour cadence; last submission HTTP 202 Accepted.

FARMSKING TECHNOLOGIES PRIVATE LIMITED

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AUTHORIZED SIGNATORY

Board of Directors • Technology Division